Breaking the Classics

Cryptography Lab Report

Zakariya Sabri · Intro to Cryptography · Python

Objective

Use Python to understand and break three classical ciphers: Caesar, substitution, and Vigenère.

Warm-up: Caesar cipher

I wrote `Encrypt()` and `Decrypt()` functions. Each letter is shifted by a key value. Modulo 26 keeps letters inside the alphabet, and spaces remain unchanged. With key **4**, the result is:

```
CRYPTO IS FUN UNTIL THE PROFESSOR SAYS QUIZ
GVCTXS MW JYR YRXMP XLI TVSJIWWSV WECW UYMD
```

I also included a loop to try all 26 possible shifts when the key is unknown. The loop is currently commented out. Caesar brute force is practical because its key space is small.

Room 1: Substitution cipher

I wrote a function to count letters and sort them from most frequent to least frequent. I combined frequencies with repeated words and English word patterns to build a substitution mapping. The recovered message was:

> The secret message is hidden in this paragraph. Letter frequency does not reveal the whole key, but it gives a good start. Look for common words, repeated patterns, and short words. When you recover the message, the Vigenere key has four letters.

The clue was that the next key had **four letters**. Frequency analysis gives useful clues, but it does not always recover the complete key. Substitution brute force is impractical because there are 26! possible keys.

Room 2: Vigenère cipher

I split the ciphertext into four groups using `cipher[0::4]`, `cipher[1::4]`, `cipher[2::4]`, and `cipher[3::4]`. Each group uses the same key letter, so it behaves like a Caesar cipher.

I assumed that each group's most frequent ciphertext letter represented plaintext **E**. This gave the key **CODE**, with shifts **2, 14, 3, 4**. I decrypted the message:

> The secret keeper left these notes because the letter E appears everywhere. Every team member should check each column, seek the most frequent symbol, and use it to recover the key. When the message becomes clear, enter the access word ECLIPSE.

Access word: ECLIPSE. The decrypted output had no spaces because the ciphertext had no spaces. The readable version was added manually.

Code notes

- The Caesar functions expect uppercase English letters and spaces.
- To display correct percentages, the frequency function must divide each count by the number of letters only, then multiply by 100.
- Remove the duplicate, unreachable `M` condition in Room 1; keep the mapping `M` to `b`.

What I learned

Caesar can be broken by brute force. Substitution can be attacked using language patterns. A repeating Vigenère key allows the ciphertext to be separated into Caesar groups and analyzed using letter frequencies.